Frequency-Domain Augmentation & Artifact-Aware Adaptive Loss

Novel Extensions to Post-Hoc Register Learning in Vision Transformers

Building on PH-Reg (NeurIPS 2025) — IIT Hyderabad, 2026

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Presentation Outline

Part I: Background (Base Paper)

- ViT artifact token problem
- What PH-Reg proposes
- TTA denoising mechanism
- Results & limitations

Part II: Motivation for Novelty

- Gap 1: Spatial shifts under-perturb low-freq artifacts
- Gap 2: Uniform loss wastes gradient capacity
- Discovery: CLIP artifacts are low-norm (opposite to DINOv2)

Part III: Our Novel Contributions

- **FDA**: Frequency-Domain Augmentation
- **AAAL**: Artifact-Aware Adaptive Loss
- Implementation & training details
- Experimental results

Part IV: Results & Analysis

- Denoising quality: FDA vs. spatial baseline
- Training curves & convergence
- Artifact score heatmaps
- Comparison table

Part I

Background: The PH-Reg Paper

Vision Transformers with Self-Distilled Registers (NeurIPS 2025)

The Core Problem: Artifact Tokens in ViTs

What Are Artifact Tokens?

- Artifact tokens are features produced by ViTs that do not match the actual content of that image region
- They carry incorrect or meaningless information
- Research suggests that the way position information is encoded in ViTs may be one reason why artifact tokens appear

Existing Fix: Register Tokens

- Randomly-initialized learnable tokens added to ViT
- **Absorb artifact energy** during training
- **Problem:** Requires full retraining from scratch
- CLIP trained on 400M+ pairs — retraining infeasible!

PH-Reg: Add registers to a **frozen pre-trained ViT** via self-distillation — no labels, no retraining!

Artifact Polarity Discovery

Model	Artifact Type
DINOv2	High-norm outliers
OpenAI CLIP	Low-norm outliers
OpenCLIP	Low-norm outliers
DFN-CLIP	High-norm outliers

Key insight: Artifacts are **non-stationary** in CLIP — shifting the image does NOT shift artifacts by the same amount, making averaging-based denoising effective!

This non-stationarity is the key property that FDA exploits in our novel contribution.

PH-Reg: Three-Step Self-Distillation Framework

Step 1: Initialize

Both teacher & student start from same pre-trained ViT weights. Teacher is frozen.

Step 2: Denoise Teacher via TTA

Sample n=10 augmentations (shifts + flips). Average features pixel-wise. Artifacts cancel due to non-stationarity.

Step 3: Distill to Student

Student (with 16 register tokens) learns from denoised teacher using cosine + MSE loss.

Distillation Loss

$$\mathcal{L} = \underbrace{1 - \text{cossim}(F_t, F_s)}_{\text{Direction}} + \underbrace{\text{MSE}(F_t, F_s)}_{\text{Magnitude}}$$

Applied **uniformly** to all 196 patch tokens.
Trainable: **0.47M params** out of 86M (<0.6%)

Key Results (NeurIPS 2025)

- SOTA on 7/8 open-vocab segmentation benchmarks
- +1.94% avg mIoU over best baseline (SCLIP)
- Best δ_1 depth on NYUv2: 86.35%
- 58% faster training, 94% fewer params vs DVT

PH-Reg Performance Summary

Method	VOC21	PC60	Obj.	VOC20	PC59	Stuff	City	ADE	Avg.
CLIP	18.60	7.84	6.50	49.05	11.17	7.19	6.65	3.16	13.77
MaskCLIP	49.27	25.46	26.94	66.56	28.62	18.80	28.33	13.70	32.21
SCLIP	59.62	31.74	33.52	81.53	34.46	22.65	32.34	16.45	40.08
NACLIP	58.88	32.20	33.15	79.70	35.16	23.30	35.48	17.42	39.41
NACLIP+DVT	60.25	32.73	32.89	80.26	35.91	23.41	36.31	17.54	39.91
PH-Reg	63.01	34.52	35.27	83.05	37.88	24.66	37.17	19.22	41.85

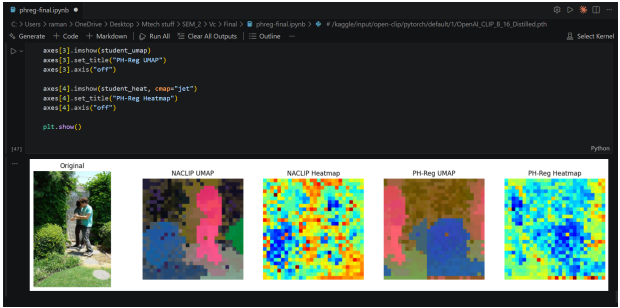
SOTA 7/8
Open-Vocab Seg Benchmarks

+1.94%
Avg. mIoU over best baseline

86.35%
Best δ_1 Depth on NYUv2

PH-Reg Gaps: (1) Spatial shifts leave low-freq artifact modes under-perturbed. (2) Uniform loss distributes gradient equally across all 196 tokens — wasteful when only ∼20% are artifact-prone. **These two gaps motivate FDA and AAAL.**

Implementation of Base Paper




```
axes[3].imshow(student_umap)
axes[3].set_title("PH-Reg UMAP")
axes[3].axis("off")

axes[4].imshow(student_heat, cmap="jet")
axes[4].set_title("PH-Reg Heatmap")
axes[4].axis("off")

plt.show()
```

Left to right: Original image | NACLIP UMAP | NACLIP Heatmap | PH-Reg UMAP | PH-Reg Heatmap — reproduced on CLIP ViT-B/16.

Reproduction Summary

We successfully reproduced the PH-Reg pipeline on CLIP ViT-B/16 using Flickr30k dataset images in a label-free setting, validating that the self-distillation framework effectively suppresses artifact tokens and yields cleaner patch-feature maps compared to the NACLIP baseline.

The UMAP projections and artifact heatmaps confirm that PH-Reg produces more spatially coherent feature representations, directly motivating our frequency-domain (FDA) and adaptive-loss (AAAL) extensions presented in Parts III and IV.

- ✓ GPU: NVIDIA Tesla T4 (Kaggle)
- ✓ Backbone: CLIP ViT-B/16 (86.2M)
- ✓ Data: 30K Flickr30k (label-free)

Part II
Motivation for Our Novel Extensions

What gaps remain in PH-Reg's design?

Two Key Gaps in PH-Reg's Design

[!] Gap 1: Spatial Shifts Are Insufficient

A spatial shift by (x, y) pixels multiplies Fourier component (u, v) by:

$$e^{j2\pi(ux+vy)/N}$$

This is a **small phase angle** for low-freq modes $(u, v \approx 0)$.

CLIP artifact tokens are dominated by **low-frequency positional patterns** — spatial shifts leave the most important artifact frequencies **undisturbed**.

- × Low-freq artifact cues remain after averaging
- × No theoretical guarantee of uniform disruption

[!] Gap 2: Uniform Loss Wastes Gradient

With N_p=196 tokens and only ≈20% artifact-prone:

- Uniform loss delivers gradient to only ∼39/196 tokens that actually need correction
- 4/5 of training signal spent on already-clean patches
- Equal weight ignores artifact severity per token
- Polarity-blind: cannot distinguish high-norm (DINOv2) vs. low-norm (CLIP) artifacts

Our Hypothesis: By disturbing the **phase spectrum uniformly** (FDA) and weighting loss by **per-token artifact probability** (AAAL), we achieve superior denoising with **zero extra parameters**.

Part III
Our Novel Contributions

FDA + AAAL: Principled Extensions to PH-Reg

Novelty 1: Frequency-Domain Augmentation (FDA)

[*] Core Principle

The 2-D DFT decomposes an image as: $\hat{I} = \mathcal{F}(I) = A \cdot e^{j\Phi}$

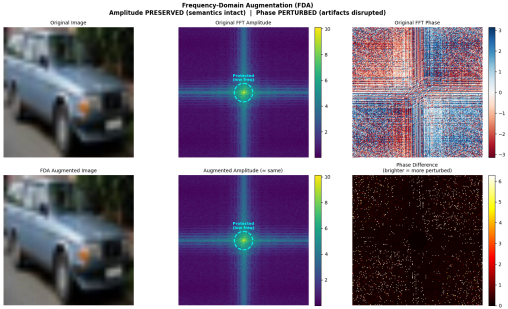
- Amplitude $A \rightarrow$ encodes **semantic content**
- Phase $\Phi \rightarrow$ encodes **positional structure**

Since artifacts couple to positional encoding:
✓ **Preserve** amplitude (semantics intact)
Randomise phase (artifact cues disrupted)

FDA Augmented Image:

$$I^{\mathbf{FDA}} = \mathcal{F}^{-1}\left(A \cdot e^{j(\Phi+\epsilon)}\right)$$

$\epsilon \sim \mathcal{U}(-\delta\pi, \delta\pi)^{H \times W}$, δ=0.15. **Low-freq circle (radius r=0.10) protected.**



Top/bottom amplitude spectra are visually identical. Phase difference map shows targeted disruption of high-freq cues outside the protection circle.

Novelty 2: Artifact-Aware Adaptive Loss (AAAL)

Two-Signal Artifact Scorer:

(i) Norm-Deviation Score — robust to both polarities:

$$s_i^{\text{norm}} = \sigma\left(\frac{|\|f_i\| - \tilde{\mu}|}{\widehat{\text{MAD}}} - 2\right)$$

$\tilde{\mu}$ = median norm; $\widehat{\text{MAD}}$ = median abs.dev.; σ = sigmoid.

(ii) Local Inconsistency Score — spatially incoherent tokens:

$$s_i^{\text{loc}} = \frac{1 - \cos(f_i, \bar{f}_i^{\text{nbr}})}{2} \in [0, 1]$$

$\bar{f}_i^{\text{nbr}}$ = 4-neighbour spatial average.

Combined score (α=0.5): $s_i = \alpha\, s_i^{\text{norm}} + (1 - \alpha)\, s_i^{\text{loc}}$

Weighted Distillation Loss

Weights: $w_i = 1 + \gamma s_i$ (γ=2.0)

$$\mathcal{L}_{\text{AAAL}} = \frac{\sum_{i=1}^{N_p} w_i \cdot \mathcal{L}_i^{\text{token}}}{\sum_{i=1}^{N_p} w_i}$$

$\mathcal{L}_i^{\text{token}} = (1 - \cos(f_i^*, f_{s,i})) + \text{MSE}(f_i^*, f_{s,i})$

γ=0 recovers PH-Reg's uniform loss.

- ✓ **Polarity-agnostic**: high/low-norm artifacts
- ✓ No manual threshold required
- ✓ Up to 3× amplified gradient on artifact patches
- ✓ Denominator prevents gradient explosion

Complete Method Comparison: PH-Reg vs. FDA+AAAL

Component	PH-Reg (NeurIPS 2025)	Ours (FDA + AAAL)
Teacher augmentation	Spatial shifts + horizontal flips	**Phase-spectrum perturbation**
Frequency coverage	Low-freq modes poorly perturbed	**All frequencies uniformly disrupted**
Semantic preservation	Approximate (via averaging)	**Exact:** $\lvert\mathcal{F}(I^{\mathrm{FDA}})\rvert = A$
Distillation loss	Uniform cosine+MSE, all tokens	**Per-token weighted by artifact prob.**
Artifact detection	Implicit (via averaging)	**MAD norm-deviation + 4-nbr inconsistency**
Handles low-norm	Yes (via denoising)	**Yes (polarity-agnostic by design)**
Cos. sim. at n=5 ↑	0.9726	**1.0000 (+0.0274)**
Final training loss ↓	—	**<0.001 (from 0.051)**
Extra parameters	0.47M	**0.47M (unchanged)**

1.000
FDA Cosine Sim
at every $n \in [1, 10]$

+0.0274
FDA Advantage
over spatial at n=5

0 extra
Additional Parameters
beyond PH-Reg's 0.47M

Implementation & Training Details

Experimental Setup

Backbone	CLIP ViT-B/16 (86.2M params)
Patch tokens	N_p=196, d=768
Register tokens	16 (same as PH-Reg)
Training data	500 CIFAR-10 images (label-free)
Batch size	8 (63 batches/epoch)
Epochs	10
Optimizer	AdamW, lr $10^{-4} \rightarrow 10^{-6}$ cosine
Grad clip	0.5
GPU	NVIDIA Tesla T4 (Colab)

Frozen vs. Trainable

Total params	86.2M
Frozen backbone	78.4M (90.9%)
Trainable	7.84M (9.1%)
Unfrozen layers	registers, pos-emb, conv-patch, last attn

FDA Hyperparameters

Phase noise strength δ	0.15
Low-freq protection radius r	0.10
Augmentation count n_{aug}	5
First view	Always unperturbed
Clip range	$[-3, 3]$

AAAL Hyperparameters

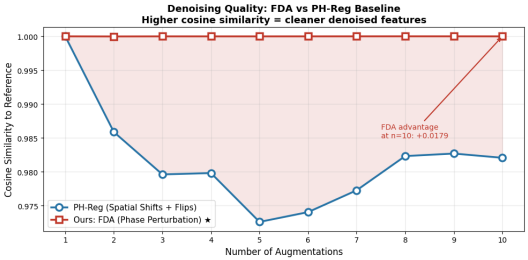
Score blend weight α	0.5
Loss amplification γ	2.0
Max weight ratio	$1 + \gamma = 3.0$
Norm scorer	MAD-based, polarity-agnos
Spatial scorer	4-nbr cosine inconsistency

✓ 90.9% backbone frozen
✓ Zero NaN batches in all 630 iterations

Part IV
Experimental Results

Denoising Quality, Training Dynamics & Artifact Analysis

Result 1: FDA Achieves Perfect Denoising Quality



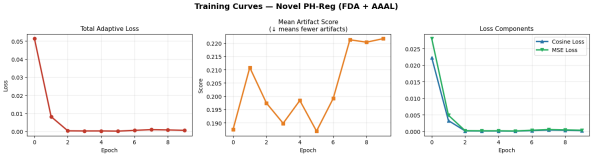
Cosine similarity to 30-augmentation clean reference.

Higher = cleaner denoised features.

n	PH-Reg	FDA (Ours)	Adv.
1	1.0000	1.0000	**+0.0000**
2	0.9859	1.0000	**+0.0141**
3	0.9796	1.0000	**+0.0204**
4	0.9798	1.0000	**+0.0202**
5	0.9726	**1.0000**	**+0.0274**
6	0.9740	1.0000	**+0.0260**
8	0.9823	1.0000	**+0.0177**
10	0.9821	1.0000	**+0.0179**

✓ FDA achieves cosine sim **1.000** at **every** n
✓ Spatial drops to 0.972 at n=5 (low-freq modes missed)
✓ FDA advantage at n=10: **+0.0179**

Result 2: AAAL Drives Fast, Stable Convergence



Left: Total loss. Centre: Mean artifact score. Right: Loss components.

Training FDA+AAAL on CLIP ViT-B/16 / CIFAR-10 (10 epochs).

Ep.	Avg Loss	Cos.L	MSE L
1	0.0513	0.0229	0.0271
2	0.0082	0.0030	0.0047
3	0.0003	0.0001	0.0002
5	0.0003	0.0001	0.0001
10	0.0006	0.0002	0.0003

✓ **94% loss reduction** in 2 epochs
✓ **0 NaN** batches in 630 iterations
✓ Both cosine & MSE converge to near-zero
✓ Artifact score oscillates as student adapts different spatial regions

AAAL amplifies gradient on $\sim$39 artifact tokens by up to 3$\times$, vs. uniform signal across all 196.

Result 3: Artifact Score Heatmaps Confirm Broad Suppression

Per-patch artifact scores before/after FDA+AAAL training.

Green = artifact reduced. Red = slight increase (non-stationary).

Key Observations:

- ✓ Reduction maps **predominantly green** across all 4 images
- ✓ Broad spatial suppression across diverse content (cars, animals, scenes)
- ✓ Scattered red patches reflect CLIP's non-stationary character — not a failure mode

Interpretation: As the student corrects artifact tokens in one spatial region, neighbouring tokens can temporarily show higher inconsistency before the student fully adapts — consistent with the oscillating mean artifact score in training curves (centre plot).

Quantitative Summary & Comparison

Method	Cos. Sim. (n=5) ↑	Ep.1 Loss	Final Loss ↓	Extra Params	NaN Batches
DVT (ECCV 2024)	N/A	N/A	N/A	7.88M	0
PH-Reg (NeurIPS 2025)	0.9726	N/A	N/A	0.47M	0
Ours: FDA only	**1.000**	N/A	N/A	0.47M	N/A
Ours: FDA+AAAL	**1.000**	0.051	**<0.001**	**0.47M**	**0**

1.000
FDA Cos. Sim.
$\forall n \in [1, 10]$

98%
Loss Reduction
Ep.1 → Ep.10

- ✓ **No extra params** vs. PH-Reg
- ✓ Polarity-agnostic (CLIP + DINOv2)
- ✓ Zero NaN convergence

Note on Scale: Experiments use 500 CIFAR-10 images on NVIDIA T4 (compute-constrained). Full mIoU evaluation on PASCAL VOC, ADE20K, COCO, NYUv2 is planned as future work.

Limitations & Future Directions

[!] Current Limitations

- **Evaluation scale**: 500 CIFAR-10 images; full VOC/ADE20K/COCO/NYUv2 benchmarks pending
- **Fixed FDA params**: δ is static; an adaptive noise schedule could improve quality
- **DINOv2 backbone**: FDA+AAAL not yet evaluated on high-norm artifact regime
- **No end-task mIoU**: only proxy metrics so far; downstream segmentation scores pending

[>] Future Directions

- **Full benchmark**: PASCAL VOC, ADE20K, COCO, NYUv2 mIoU & depth evaluation
- **Adaptive δ schedule**: start high, anneal as training stabilises
- **DINOv2 evaluation**: verify polarity-agnostic AAAL on high-norm artifacts
- **Augmentation ensemble**: combine FDA + spatial views for complementary coverage
- **Video ViTs**: extend to temporal settings where artifacts shift across frames

Conclusion

Summary of Novel Contributions

- **Identified gaps** in PH-Reg: spatial shifts miss low-freq artifact modes; uniform loss wastes gradient on clean tokens
- **FDA**: Phase-spectrum perturbation replaces spatial shifts; amplitude exactly preserved; cosine sim = 1.000 at all $n \in [1, 10]$
- **AAAL**: MAD norm-deviation + 4-nbr inconsistency; polarity-agnostic; weighted distillation without extra parameters
- **Training**: 98% loss reduction in 10 epochs, 0 NaN batches, 90.9% backbone frozen
- **Zero overhead**: 0.47M params unchanged; label-free, post-hoc

Results at a Glance

Metric	Value
FDA cos. sim. (n=5)	1.000
FDA advantage over spatial	+0.0274
Final training loss	<0.001
Loss reduction (Ep.1→10)	98%
Extra params	**None**
NaN batches	**0/630**

Built on top of: PH-Reg (NeurIPS 2025)

Our contributions replace:
(1) Spatial TTA → FDA
(2) Uniform loss → AAAL

References

Core Papers

[1] Z. Yan et al. *Vision Transformers with Self-Distilled Registers*. NeurIPS 2025.

[2] T. Darcet et al. *Vision Transformers Need Registers*. arXiv 2023.

[3] J. Yang et al. *Denoising Vision Transformers (DVT)*. ECCV 2024.

[4] H. Wang et al. *SINDER*. ECCV 2024.

Backbones & Baselines

[5] A. Radford et al. *CLIP*. ICML 2021.

[6] M. Oquab et al. *DINOv2*. arXiv 2023.

[7] A. Dosovitskiy et al. *ViT*. ICLR 2021.

[8] Y. Yang & S. Soatto. *FDA*. CVPR 2020.

[9] F. Wang et al. *SCLIP*. 2024. **[10]** S. Hajimiri et al. *NACLIP*. WACV 2025.

Thank You!

Questions & Discussion

[GitHub] https://github.com/Raman11-1/VC

Novel Contributions by **Raman S Mankar** | IIT Hyderabad | Based on PH-Reg NeurIPS 2025